

# Embedding#

<https://pytorch.org/docs/stable/generated/torch.nn.quantized.Embedding.html>

## Description:

A quantized Embedding module with quantized packed weights as inputs. We adopt the same interface as `torch.nn.Embedding`, please see <https://pytorch.org/docs/stable/generated/torch.nn.Embedding.html> for documentation. Similar to `Embedding`, attributes will be randomly initialized at module creation time and will be overwritten later weight (Tensor) – the non-learnable quantized weights of the module of shape  $(\text{num\_embeddings}, \text{embedding\_dim})$ . Create a quantized embedding module from a float module `mod` (Module) – a float module, either produced by `torch.nn.quantization` utilities or provided by user

## Function Signature

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```
class torch.nn.quantized.Embedding(num_embeddings,  
embedding_dim, padding_idx=None, max_norm=None,  
norm_type=2.0, scale_grad_by_freq=False, sparse=False,  
_weight=None, dtype=torch.qint8)[source]#
```

Variables

Examples::

```
classmethod from_float(mod,  
use_precomputed_fake_quant=False)[source]#
```

Parameters

## Code Examples

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```
>>> m = nn.quantized.Embedding(num_embeddings=10,  
embedding_dim=12)  
>>> indices = torch.tensor([9, 6, 5, 7, 8, 8, 9, 2, 8])  
>>> output = m(indices)  
>>> print(output.size())  
torch.Size([9, 12])
```

# Detailed Documentation

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## Documentation

A quantized Embedding module with quantized packed weights as inputs. We adopt the same interface as `torch.nn.Embedding`, please see <https://pytorch.org/docs/stable/generated/torch.nn.Embedding.html> for documentation.

Similar to Embedding, attributes will be randomly initialized at module creation time and will be overwritten later

`weight` (Tensor) – the non-learnable quantized weights of the module of shape  $(\text{num\_embeddings}, \text{embedding\_dim})$ .

Create a quantized embedding module from a float module

`mod` (Module) – a float module, either produced by `torch.ao.quantization` utilities or provided by user